MBBS IV Junior Clerkship Whole Class Session

Premature puberty: Puberty and related disorders

Dr Grace Poon
Consultant Paediatrician
Queen Mary Hospital

Puberty and related disorders

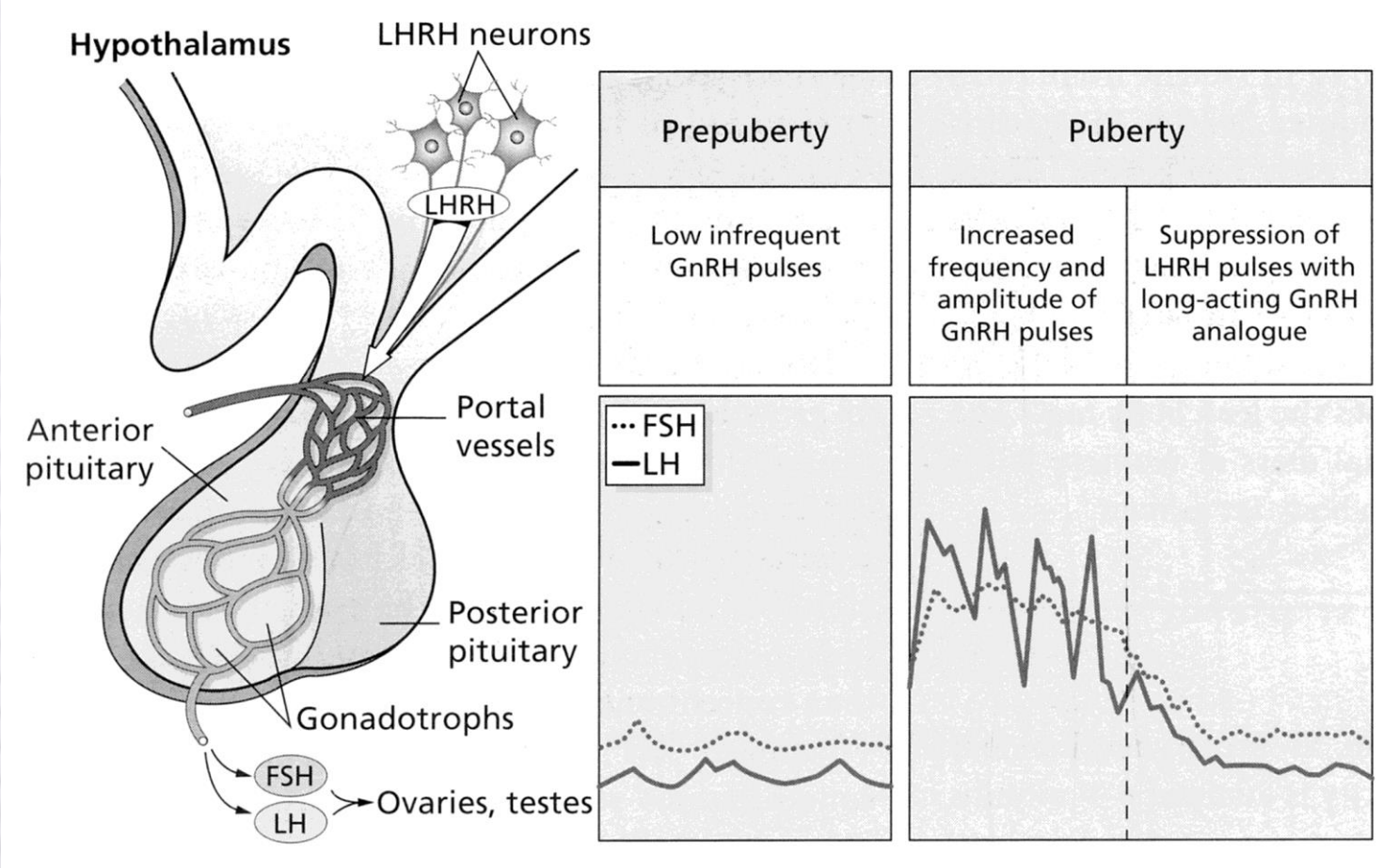
- Normal Pubertal development
- Definition of precocious puberty
- Variations of normal pubertal development
- Causes of precocious puberty
- Management of precocious puberty

Normal pubertal development

Normal pubertal development

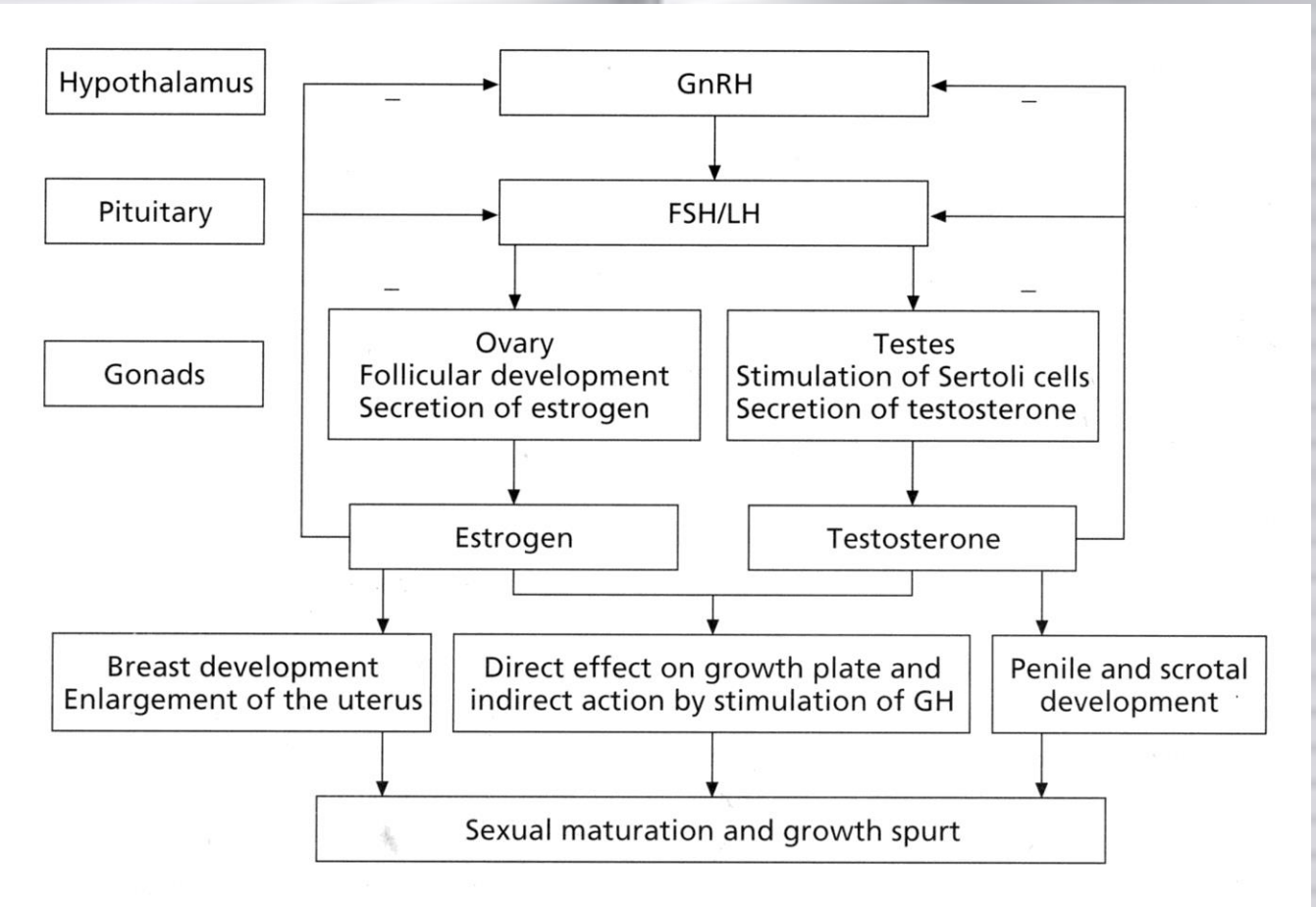
- Kisspeptins from hypothalamus bind to a G protein-coupled receptor called *GPR54* on the hypothalamic neurons to secrete GnRH
- Puberty onset: central process with activation of **GnRH pulse generator**
- Episodic ⇧ hypothalamic GnRH
  - ➔ anterior pituitary to secrete pulses of LH and FSH
  - ➔ ovaries and testes to produce oestrogen and testosterone respectively
- Gonadotropin release is controlled by GnRH and selective negative feedback inhibition of gonadal steroids at pituitary and hypothalamic levels

Hypothalamic-pituitary-gonadal axis



Brook Charles GD and Brown Rosalind S. Handbook of Clinical Pediatric Endocrinology. Blackwell Publishing, 2008, p 64.

Normal pubertal development



Brook Charles GD and Brown Rosalind S. Handbook of Clinical Pediatric Endocrinology. Blackwell Publishing, 2008, p 65.

Normal pubertal development

Onset of puberty is also modulated by:

- Genetic factors
- Peripheral signs - intrauterine and postnatal growth, BMI and fat mass, insulin sensitivity, gonadal steroid levels
- Environmental signals - light, stress, environmental endocrine disruptors

1. Sultan C (ed): Pediatric and Adolescent Gynaecology. Evidence-Based Clinical Practice. 2nd, revised and extended edition. Endocr Dev. Basel, Karger, 2012, vol 22, p 84-100.
2. Parents AS et al. Endocr Rev 2003;24:668-693.

Genes Playing a Role in Pubertal Development

Gene	Gene Product	Pubertal Disorder Associated with Gene Defect
FGFR1	Fibroblast growth factor receptor 1	Kallmann syndrome
FSH-β	FSH-β subunit	Idiopathic hypogonadotropic hypogonadism
FSHR	FSH receptor	Idiopathic hypogonadotropic hypogonadism
GNAS	$G_s\alpha$ protein	Precocious puberty (activating mutation) in McCune-Albright syndrome
GNRHR	GnRH receptor	Idiopathic hypogonadotropic hypogonadism
GPR54	Kisspeptin receptor	Idiopathic hypogonadotropic hypogonadism
HESX-1	"Homeobox gene expressed in ES cells" HESX1 protein	Abnormal hypothalamic-pituitary development
KAL1	Anosmin-1	Kallmann syndrome
KISS-1	Kisspeptin	
LEP	Leptin	Idiopathic hypogonadotropic hypogonadism and obesity
LEPR	Leptin receptor	Idiopathic hypogonadotropic hypogonadism and obesity
LH-β	LH-β subunit	Idiopathic hypogonadotropic hypogonadism
LHR	LH receptor	Idiopathic hypogonadotropic hypogonadism (inactivating mutation); "testotoxicosis" (activating mutation)
LHX3	"LIM family of homeobox transcription factors"	Abnormal hypothalamic-pituitary development
NELF	Nasal embryonic luteinizing hormone-releasing hormone factor	Kallmann syndrome
NR5A1 (SF-1)	Nuclear receptor, subfamily 5, group A, member 1 (Steroidogenic factor-1)	Abnormal hypothalamic-pituitary development, adrenal hypoplasia
NR0B1 (DAX-1)	Nuclear receptor, subfamily 0, group B, member 1 (dosage-sensitive sex reversal—Adrenal hypoplasia congenital critical region on the X chromosome)	Abnormal hypothalamic-pituitary development, adrenal hypoplasia
PC1	Prohormone convertase-1	Idiopathic hypogonadotropic hypogonadism and hypoglycemia
PROK2	Prokineticin-2	Kallmann syndrome
PROKR2	Prokineticin receptor 2	Kallmann syndrome
PROP1	"Prophet of Pit-1"	Abnormal hypothalamic-pituitary development

Nakamoto JM, Franklin SL and Geffner ME. Pediatric Practice Endocrinology. McGraw-Hill, 2010, p 260.

Normal pubertal development in girls

- Puberty starts in both sexes around 11 years of age
- More evident in girls because of breast development and the onset of growth spurt
- Breast development is caused by oestrogen and may be unilateral for several months
- Pubic and axillary hair growth under influence of adrenal androgens but the stage of breast development usually correlates well with the stage of pubic hair development

Normal pubertal development in girls

- Growth accelerates as breast development starts and peak height velocity is attained 6-9 months after breast stage 2
- Menarche occurs 2-2.5 years after breast budding
- The first menstrual bleedings are anovulatory
- Time from menarche to regular cycling is about 3 years
- The average girl will grow for another 6-7cm after menarche

Mean ages for normal pubertal development stages in girls, based on Tanner stages for breast development (B) and pubic hair (P)

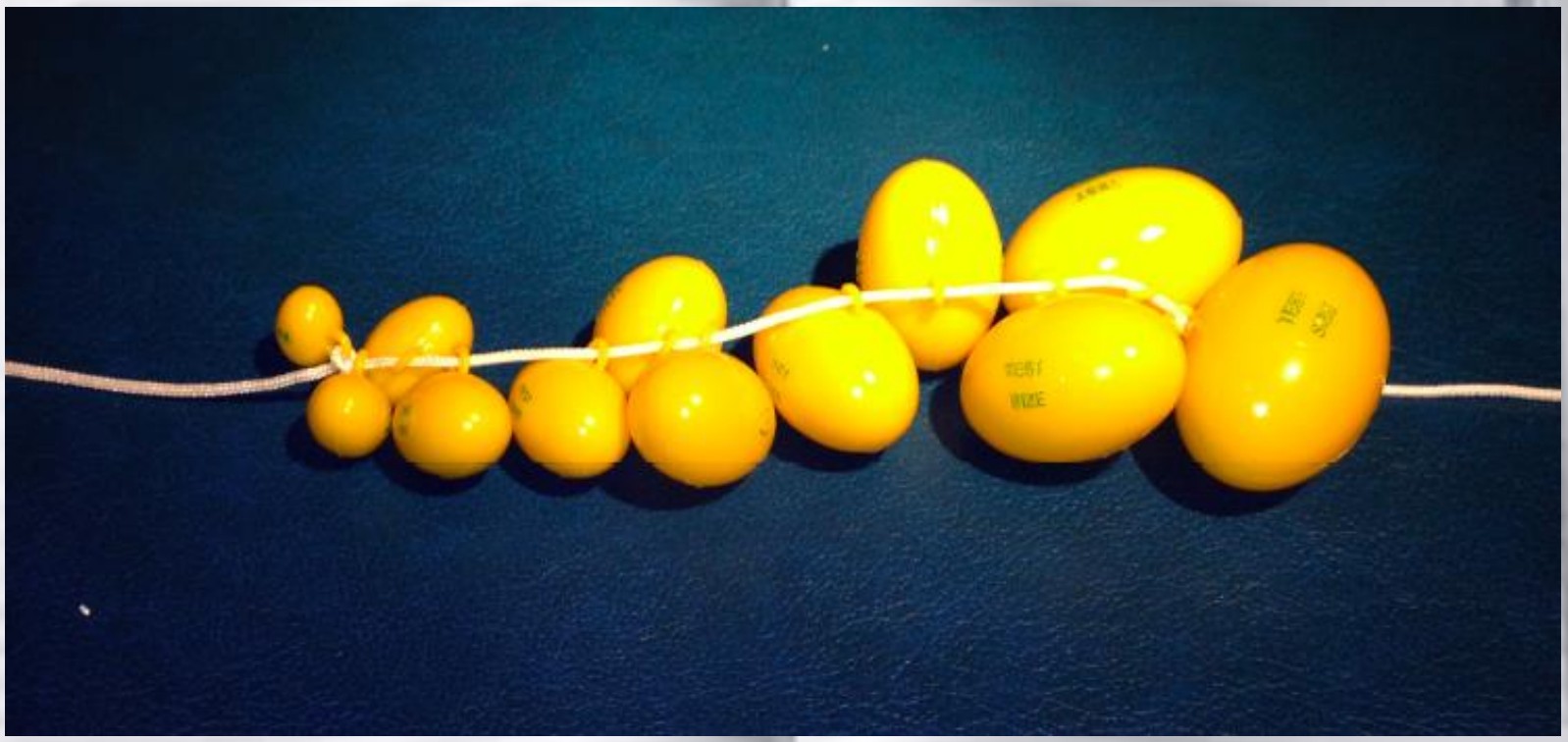
Pubertal stage	Age, years
Breast buds (B2)	10.1
Sparse pubic hair growth (P2)	11.2
Darker, coarser pubic hair growth (P3)	12.2
Growth spurt	12.2
Menarche	12.7
Adult pubic hair in type and quantity (P5)	14
Mature breast (B5)	14

Sultan C (ed): Pediatric and Adolescent Gynaecology. Evidence-Based Clinical Practice. 2nd, revised and extended edition. Endocr Dev. Basel, Karger, 2012, vol 22, p 84-100.

Normal pubertal development in boys

- 1st sign of puberty is ⇧in testicular volume to 4 ml, due to enlargement of Sertoli cells rather than Leydig cells
- Growth of penis and genitalia correlate well with pubic hair development as both are regulated by androgen
- Spermatogenesis histologically from 11-15 years
- Sperm found in early morning urine at mean of 13.3 years
- Ejaculation occurs by a mean of 13.5 years
- Adult morphology, motility and concentration of sperm are not found until bone age advances to 17 years but immature-appearing boys can be fertile

Prader orchidometer for determination of testicular volume



Normal pubertal development in boys

- Boys have peak growth spurt during mid-puberty, at mean 13.5 years, corresponding to testicular volume of 10-12 ml
- Breast enlargement common during early puberty because of ⇧oestrogen production by aromatization of testosterone before testosterone achieves concentrations that can oppose the oestrogen
- Breast tissue often regresses within 2 years
- Occasionally gynaecomastia remains permanent in normal, often obese, boys and frequently in pathological conditions such as Klinefelter syndrome or partial androgen resistance, in which the effective amount of bioactive testosterone is reduced

Definition of precocious puberty

Defining precocious puberty

- Onset of puberty at a younger age than expected for the normal population (i.e. > 2 SDs earlier than population mean)
- Classically, pubertal signs occurring before 8 years for girls or 9 years for boys have been considered precocious

Defining precocious puberty

Data based on a cross-sectional observational study of >17000 girls from the Pediatric Research in Office Setting (PROS) network suggested that puberty may be occurring earlier than previously thought:

- White girls with thelarche at a mean of 9.96 ± 1.82 years
- Black girls with thelarche at a mean of 8.87 ± 1.93 years
- 6.7% of white and 27.2% of black girls from this study would have been characterized as having precocious puberty if using the historic cutoff of 8 years

Herman-Giddens ME et al. Pediatrics 1997;99:505-512.

Defining precocious puberty

- LWPES (now the Pediatric Endocrine Society) recommended lowering age cutoff for precocious puberty to < 7 yrs in white girls and < 6 yrs in black girls
- Pubertal onset occurring between 7-8 yrs in white girls and 6-8 yrs in black girls should be evaluated if there is:
  - unusual rapid pubertal progression resulting in bone age advancement by > 2 yrs and a predicted height that is < 150cm or 2 SDs below genetic target height
  - signs and symptoms of a CNS lesion, or
  - behaviour-based factors suggesting a child's emotional state is adversely affected by early puberty and potential early menses

Kaplowitz PB, Oberfield SE. Pediatrics 1999;104:936-941.

Defining precocious puberty

- New guidelines not universally accepted by endocrinologists and several have questioned the validity of the PROS study
- Sexual development in girls younger than 8 years and boys younger than 9 years need a thorough history and examination, a bone age and close longitudinal follow up
- Necessity for and type of treatment will usually require input from specialist
- Not all organic aetiologies and certainly not the normal variants require treatment

Onset of Breast and Pubic Hair Development and Menses in Urban Chinese Girls

RESULTS: Data were analyzed for 20 654 apparently healthy girls. At age 8 years, 19.57% of these girls had evidence of breast development. The median ages of onset of Tanner stages 2 and 3 for breast development were 9.20 (95% CI: 9.06–9.32) years and 10.37 (95% CI: 10.28–10.45) years, respectively. The median ages of onset of Tanner stages 2 and 3 for pubic hair development were 11.16 (95% CI: 11.03–11.29) years and 12.40 (95% CI: 12.25–12.55) years, respectively. Menses occurred at 12.27 years (95% CI: 12.16–12.39).

CONCLUSIONS: These data suggest that urban Chinese girls are actually experiencing earlier breast development than currently used norms. The up-to-date reference for normal pubertal development in urban Chinese girls needs to be established for the purpose of determining precocious puberty or pubertal delay. *Pediatrics* 2009;124: e269–e277

Pubertal development timing in urban Chinese boys

H.-M. Ma,* S.-K. Chen,† R.-M. Chen,‡ C. Zhu,§ F. Xiong,¶ T. Li,** W. Wang,†† G.-L. Liu,‡‡ X.-P. Luo,§§ L. Liu,¶¶ M.-L. Du* and Pubertal Study Group of the Society of Pediatric Endocrinology and Genetic Disease, Chinese Medical Association[1]

We describe current pubertal development in healthy urban Chinese boys. A cross-sectional study of the pubertal development of 18 807 urban Chinese boys aged from 3.50 to 18.49 years was conducted between 2003 and 2005. Testicular volume was evaluated with a Prader orchidometer. Pubic hair development was assessed according to the Tanner method. Data on spermarche were collected using the status quo method. Probit analysis was used to calculate the median age and 95% CI at different stages of testicular development, pubic hair development and spermarche. By age 9, 12.99% of the boys had a testicular volume of 4 mL or greater. The median age of onset of puberty defined as the age at attainment of testicular volume of 4 mL or greater was 10.55 (95% CI 10.27–10.79) years. The median age for onset of pubic hair development (PH_2) and spermarche was 12.78 (95%CI 12.67–12.89) years and 14.05 (95%CI 13.80–14.32) years, respectively. Pubertal onset in urban Chinese boys is earlier than currently used clinical norms but their pubic hair development occurs relatively late in comparison with the reported data from numerous other countries. There is also evidence of a secular trend towards an earlier age of spermarche since 1979 in Chinese urban boys.

International Journal of Andrology © 2011 European Academy of Andrology, **34**, e435–e445

Not all presentations of precocious puberty are true precocious puberty

- 50-60% only one secondary sexual characteristic shows premature development
  - Isolated premature thelarche
  - Isolated premature adrenarche
  - Isolated premature menarche

  Aetiology should be sought and need monitoring to prevent precocious onset of puberty

Precocious puberty

- Only 10-20% of presentations of precocious puberty are central precocious puberty (CPP)
  - **Must exclude CNS tumour, particularly in boys**
  - Progresses rapidly and the accelerated bone maturation causes early fusion of bone plates and compromising final adult height
- 10% adrenal or ovarian cause
  - With autonomous hyperproduction of oestrogen resulting in precocious pseudopuberty independent of gonadotropin activation

Precocious puberty

- It has become increasingly evident that the hyperoestrogenism may have an exogenous cause, such as environmental chemical pollutants in the air, water, and food chain
- These xenoestrogens have a chemical structure that mimics the actions of natural oestrogens by stimulating the activity of target tissues

Variations of normal pubertal development

Premature thelarche

- Isolated breast development in girls between 2-7 yrs
- This differs from the genital crisis of the newborn whose breast development (associated with strong oestrogenization and even milk production) may last for the first 18 months of life
- Bilateral in half the cases, unilateral, or less frequently asymmetric
- Volume varies: 60% at B2, 30% at B3, 10% at B4
- Breast often tender and palpation sometimes painful
- No discharge

Premature thelarche

- If persistent or marked thelarche → LHRH test
- LHRH test – predominant FSH response
- Bone maturation rarely accelerated
- Progression characterized by fluctuation over time: spontaneous remission, persistence, and aggravation of breast volume, which should evoke the possibility of pubertal onset
- 30-60% regress within an average of 18 months
- In the usual form, premature thelarche does not require any treatment

Premature adrenarche/pubarche

- The appearance of pubic hair before 8 years
- More commonly in girls than in boys
- Usually isolated but occasionally axillary hair observed
- Clinical exam to detect other signs of hyperandrogenism: acne, abnormal perspiration, clitoral hypertrophy
- Growth velocity and bone age maturation are usually only slightly accelerated

Premature adrenarche/pubarche

- An androgen work-up (testosterone, 17OHP, DHEAS) +/- a synacthen test to exclude **congenital adrenal hyperplasia**
- In most situations, the maturation of the adrenal function that precedes puberty is accelerated or early, although the reason remains unknown

Premature adrenarche/pubarche

- Exaggerated adrenarche is an extreme type of premature adrenarche
  - Girls with subtle androgen excess (e.g., significant BA advancement but not clitoromegaly) or insulin resistance (e.g., central adiposity or acanthosis nigricans)
  - Often with slightly advanced onset of true puberty, but height potential not compromised
  - Adrenal steroid levels in mid- or late pubertal range
  - Testosterone not exceeding lower end of adult female range

Premature adrenarche/pubarche

- Still unclear whether a normal variant due to advanced onset of normal zona reticularis development or an early manifestation of the steroidogenic dysregulation of polycystic ovary syndrome (PCOS) with ⇧ cardiometabolic risk factors
- Premature pubarche carries a 15-20% risk of developing PCOS, esp in those with exaggerated adrenarche
- Girls with premature adrenarche should be followed through puberty

Isolated premature menarche

- Poorly defined and understood
- Menstruation without other signs of puberty in a young girl before 9 years of age
- Aetiology often unknown
- Must exclude foreign body, local masses, and McCune-Albright syndrome
- Transient ovarian activity has been observed in isolated premature menarche

Isolated premature menarche

- Several clinical manifestations: menstruation that is isolated or associated with breast development, with inconsistent response to LHRH test and the frequent presence of functional ovarian cysts
- Menstruation may recur cyclically
- These rises in oestrogenisation increase the risk of accelerated bone maturation
- Treatment to stop puberty is often discussed but rarely undertaken

Abnormal pattern of gonadotropin secretion

Primary hypothyroidism

- TSH is a weak agonist at FSH receptor resulting in ovarian stimulation with isolated breast development in girls
- In boys, testicular enlargement without other secondary sexual characteristics may occur
- No pubertal progression in majority of cases
- Delayed bone age with poor growth velocity
- Excellent prognosis with reversal of puberty once treatment is started but final height may be affected if diagnosis is delayed or if normal puberty progresses at an early age

Pathological causes of precocious puberty

Central precocious puberty

- Isosexual pubertal development with pubertal changes appropriate for the sex of the child
  - Sex budding in girls
  - Testicular enlargement in boys
- Results from activation of HPG axis at an earlier age
- At least 80% idiopathic in girls vs 10% idiopathic in boys
- Need to exclude an underlying CNS lesion
- Most concerning causes are tumours arising in the suprasellar region, with other or without (usually anterior) pituitary hormonal deficiencies
- Most common cause of CPP in the very young child is benign hypothalamic hamartoma

Etiologies of Isosexual Central Precocious Puberty

- Idiopathic
- Hypothalamic hamartomas
- Other brain tumors (glial cell tumors, germ cell tumors, and, occasionally, craniopharyngiomas and pinealomas)
- "Idiopathic" isolated GH deficiency or multiple anterior pituitary hormone deficiencies, and ectopic neurohypophysis
- Congenital suprasellar brain defects (subarachnoid cysts, arachnoidoceles, and Rathke cleft cysts)
- Congenital mid-line anomalies (hydrocephalus, meningomyelocele, and optic nerve hypoplasia)
- Cranial irradiation
- Previous meningoencephalitis
- Major head trauma
- Perinatal insults
- Neurofibromatosis-1
- Untreated or undertreated peripheral causes of puberty (eg, congenital adrenal hyperplasia)
- "Overlap syndrome," in association with longstanding untreated primary hypothyroidism
- Hyperprolactinemia
- Adoption from abroad
- Activating mutation of the gene encoding *GPR54*, the G protein-coupled receptor mediating the effects of kisspeptin on stimulation of GnRH neurons

Nakamoto JM, Franklin SL and Geffner ME. Pediatric Practice Endocrinology. McGraw-Hill, 2010, p 269.

Precocious pseudopuberty

- Gonadal or adrenal sex steroid secretion not resulting from activation of HPG axis, i.e. pituitary independent or peripheral in origin
- Loss of normal feedback and sex steroid concentrations can be very high with low gonadotropins
- Can involve isolated androgen, isolated oestrogen, or combined androgen and oestrogen production

Causes of Precocious Pseudopuberty

- Androgen overproduction: girls
  - Ovarian arrhenoblastomas
  - Ovarian hyperthecosis
- Androgen overproduction: boys
  - Leydig-cell tumors
  - hCG-secreting tumors (eg, hepatoblastomas and germ cell tumors sometimes seen in boys with Klinefelter syndrome)
  - McCune-Albright syndrome
  - Familial testotoxicosis
- Androgen overproduction: girls and boys
  - Nonclassical (late-onset) adrenal hyperplasia
  - Androgen-secreting adrenal tumors
  - Generalized glucocorticoid resistance
  - Some cases of Cushing syndrome
  - Exogenous anabolic steroid abuse
- Estrogen overproduction: girls
  - Ovarian cysts
  - Ovarian granulosa cell tumors
  - Sertoli-Leydig cell tumors (isolated or as part of Peutz-Jeghers syndrome)
  - McCune-Albright syndrome
- Estrogen overproduction: boys
  - Certain adrenal tumors
  - Sertoli cell tumors (usually in the setting of Peutz-Jeghers syndrome)

Nakamoto JM, Franklin SL and Geffner ME. Pediatric Practice Endocrinology. McGraw-Hill, 2010, p 274

Causes of Precocious Pseudopuberty

- Androgen overproduction: girls
  - Ovarian arrhenoblastomas
  - Ovarian hyperthecosis
- Androgen overproduction: boys
  - Leydig-cell tumors
  - hCG-secreting tumors (eg, hepatoblastomas and germ cell tumors sometimes seen in boys with Klinefelter syndrome)
  - McCune-Albright syndrome
  - Familial testotoxicosis
- Androgen overproduction: girls and boys
  - Nonclassical (late-onset) adrenal hyperplasia
  - Androgen-secreting adrenal tumors
  - Generalized glucocorticoid resistance
  - Some cases of Cushing syndrome
  - Exogenous anabolic steroid abuse
- Estrogen overproduction: girls
  - Ovarian cysts
  - Ovarian granulosa cell tumors
  - Sertoli-Leydig cell tumors (isolated or as part of Peutz-Jeghers syndrome)
  - McCune-Albright syndrome
- Estrogen overproduction: boys
  - Certain adrenal tumors
  - Sertoli cell tumors (usually in the setting of Peutz-Jeghers syndrome)

Nakamoto JM, Franklin SL and Geffner ME. Pediatric Practice Endocrinology. McGraw-Hill, 2010, p 274

McCune-Albright syndrome

- Multisystem disorder of both boys and girls but much more commonly in girls
- Characterised by triad of:
  - Irregularly edged hyperpigmented macules or café au lait spots “coast of Maine”
  - Polyostotic fibrous dysplasia
  - Multiple autonomous endocrinopathies; most commonly gonadotropin-independent sexual precocity; other endocrine involvement includes the thyroid, the adrenals, the pituitary and the parathyroids

At least 2 of these features should be present for the diagnosis to be made

McCune-Albright syndrome

Cafe-au-lait skin macules
with characteristic features, including jagged "coast of Maine" borders, and location respecting the midline of the body.

McCune-Albright syndrome

McCune-Albright syndrome

- Sporadic condition due to a somatic activating missense mutation in the gene encoding the α-subunit of the G-protein that stimulates cAMP production
- The mutation occurs early in embryogenesis (somatic or postzygotic, rather than germ cell in origin) and results in failure of phosphorylation of GTP to GDP and thereby constitutive activation of adenylyl cyclase in multiple affected tissues
- Such mutations may also occur in the skin, bones, liver (cholestasis), heart (arrhythmias) and GI tract

McCune-Albright syndrome

- Sexual precocity in girls is caused by autonomously functioning luteinized follicular cysts of the ovaries
- Multiple follicular cysts with an occasional large solitary cyst may be present
- Oestrogen production is associated with a prepubertal pattern of LH secretion with absent LH response to GnRH
- Later GnRH-dependent puberty ensues with ovulatory cycles
- Sexual precocity is rare in boys with MAS; when occurs, it is associated with asymmetric enlargement of the testes in addition to signs of sexual precocity

Causes of Precocious Pseudopuberty

- Androgen overproduction: girls
  - Ovarian arrhenoblastomas
  - Ovarian hyperthecosis
- Androgen overproduction: boys
  - Leydig-cell tumors
  - hCG-secreting tumors (eg, hepatoblastomas and germ cell tumors sometimes seen in boys with Klinefelter syndrome)
  - McCune-Albright syndrome
  - Familial testotoxicosis
- Androgen overproduction: girls and boys
  - Nonclassical (late-onset) adrenal hyperplasia
  - Androgen-secreting adrenal tumors
  - Generalized glucocorticoid resistance
  - Some cases of Cushing syndrome
  - Exogenous anabolic steroid abuse
- Estrogen overproduction: girls
  - Ovarian cysts
  - Ovarian granulosa cell tumors
  - Sertoli-Leydig cell tumors (isolated or as part of Peutz-Jeghers syndrome)
  - McCune-Albright syndrome
- Estrogen overproduction: boys
  - Certain adrenal tumors
  - Sertoli cell tumors (usually in the setting of Peutz-Jeghers syndrome)

Nakamoto JM, Franklin SL and Geffner ME. Pediatric Practice Endocrinology. McGraw-Hill, 2010, p 274

Familial testotoxicosis

- Rare familial condition
- Gonadotropin-independent precocious puberty only in boys, often presenting at 2-5 years with accelerated growth, early secondary sexual characteristics and reduced adult height
- Testes show only little increase in size
- Biopsy shows Leydig cell hyperplasia
- Treatment with androgen-receptor-blocking agents (ketoconazole) or aromatase inhibitors

Familial testotoxicosis or Familial male-limited precocious puberty (FMPP)

- Autosomal dominant inheritance only in males
- Associated with a number of constitutively activating mutations of LH receptor, mostly in the transmembrane domain of the receptor
- FMPP is associated with premature Leydig and germinal cell maturation
- Virilization with very high concentrations of testosterone and enlargement of testes to the early of mid-pubertal range, although seem smaller than expected in relation to stage of penile growth

Familial male-limited precocious puberty (FMPP)

- Extensive laboratory evaluation localized the source of testosterone secretion to the testes
- Testicular biopsy revealed premature maturation of Leydig cells and spermatogenic elements
- Unstimulated gonadotropin concentrations are prepubertal with a minimal response to to GnRH stimulation
- Lack of the usual pubertal pattern of LH pulsatility
- Fertility and a normal pattern of LH secretion and response to GnRH are demonstrated in adulthood

Management of precocious puberty

Evaluation of abnormal puberty

- Onset of puberty before 8 years?
- Any progression of clinical symptoms?
- Any biological or radiographic signs of exaggerated maturation?
- Is the predicted adult height affected?
- Any psychological consequences?
- Gonadotropin-dependent or independent?
- If CPP, is it due to a tumour or idiopathic?

History

- Birth history – SGA (more prone to precocious pubarche, earlier onset of pubertal development and menarche and faster progression of puberty than children born AGA)
- Past medical history: past x-rays, brain trauma, neonatal infection of the CNS
- Family history
- Precise timing and sequence of pubertal milestones
- Impact on pscychological health or well-being

Examination

- Focusing on the neurologic and endocrine systems
- Thyroid examination
- Stages of pubertal development – Tanner stages
- Neurological exam, optic fundi, visual field
- Neurocutaneous markers
- Scoliosis or body asymmetry
- Assessment of growth chart

Tanner stages of pubertal maturation

Table 1. Tanner Stages of Pubertal Maturation

Stage	Breast	Pubic Hair in Males and Females	Male Genitalia
1	Prepubertal	Prepubertal	Prepubertal
2	Breast budding, elevation of both breast and nipple as a small mound	Sparse growth of long, lightly pigmented hairs at the base of the penis in males or the mons veneris/labia majora in females	Enlargement of the testes and scrotum, thinning and reddening of the scrotal skin, penis remains prepubertal
3	Continued enlargement of both breast and areola without separation of their contours	Additional darkening and coarsening of hair, spreading over the pubic symphysis	Further growth of testes and scrotum, enlargement of the penis in length
4	The areola and nipple form a secondary mound projecting above the contour of the breast	Adult in character but confined to the suprapubic area in males and females	Further growth of testes and scrotum with pigmentation of the scrotal skin, further enlargement of the penis in width, with maturation of the glans
5	Adult shape, areola and nipple recessed to the contour of the breast	Adult in distribution with spread to the medial thighs in males and females. May extend to the lower abdomen in males.	Testes, scrotum, and penis are adult in size and shape

Fuqua JS. JCEM 2013;98:2198-2207.

Findings suggestive of pathology

- A rapid tempo of progression
- Advanced development
- Rapid linear growth
- Advanced skeletal maturation

Investigations

- Hormonal profile: LH, FSH, oestradiol / testosterone, TFT, prolactin and β-hCG
- X-ray for bone age
- LHRH test
- Pelvic USG
- MRI brain and pituitary gland

Investigations

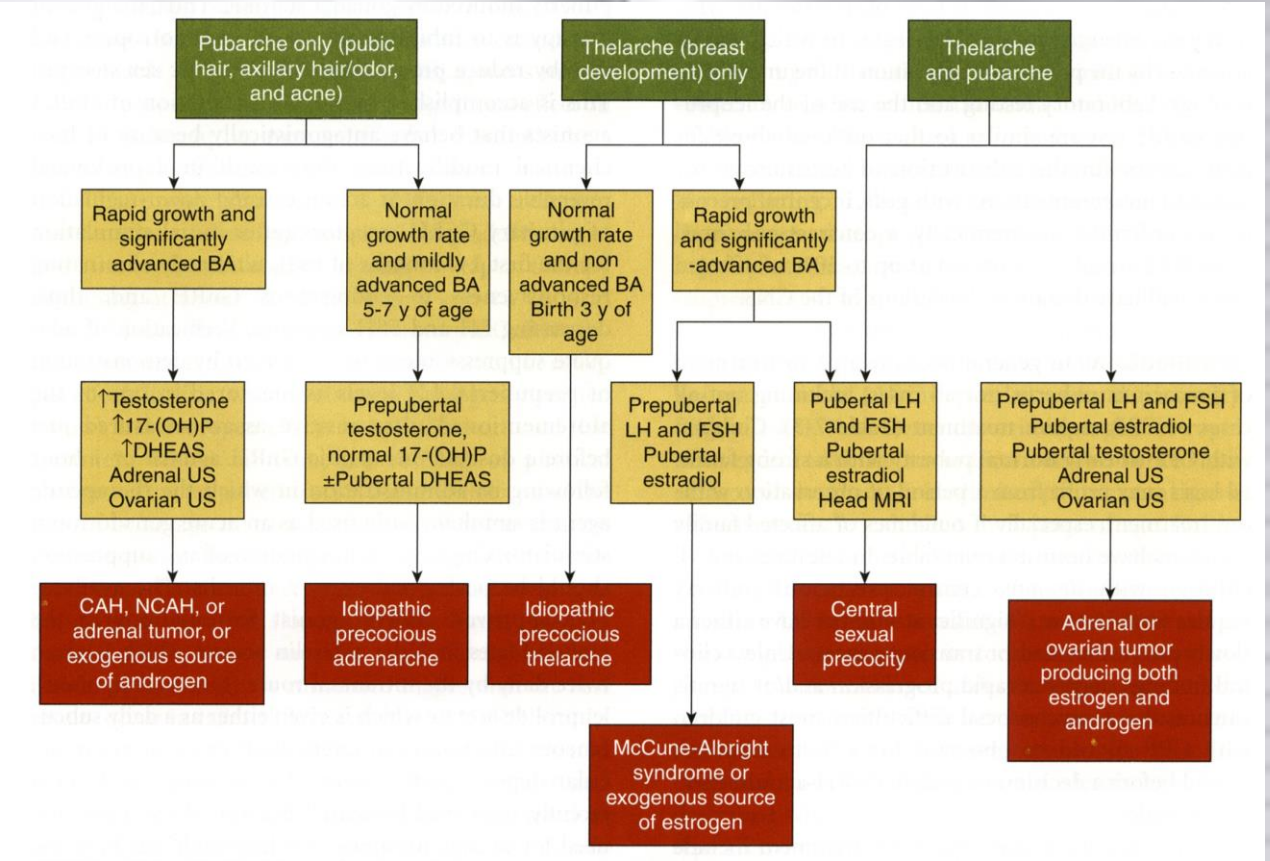
If peripheral cause is suspected:

- USG pelvis and adrenals
- Measurement of 17OHP, DHEAS and testosterone especially in girls with virilization to screen for non-classical congenital adrenal hyperplasia
- Occasionally ACTH stimulation test
- 24 hour urine free cortisol if suspicious of Cushing syndrome

Investigations

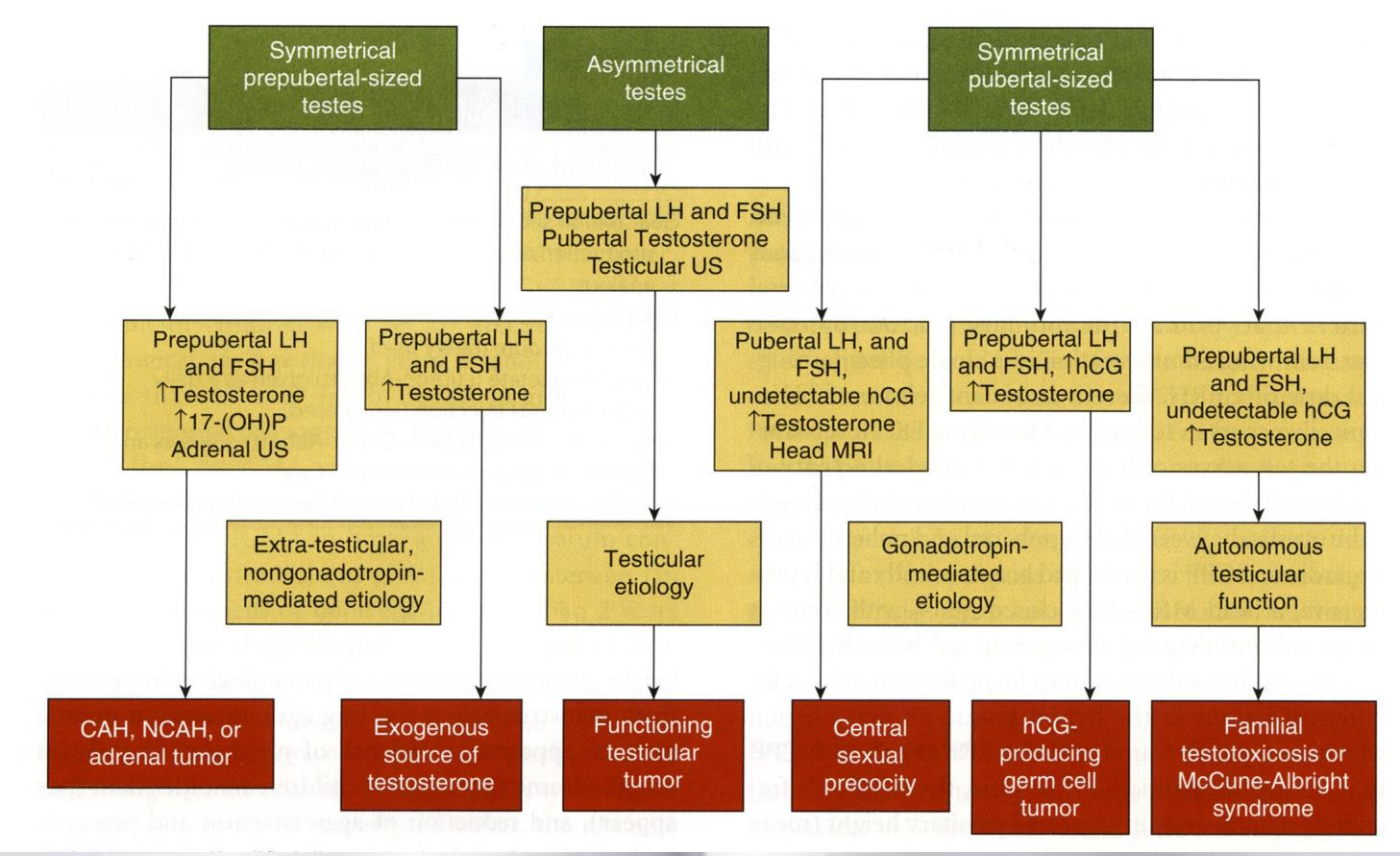
- In boys
  - If only one testis is enlarged, a radiological search for an androgen-producing tumour within the larger testis is mandatory
  - If both testes are enlarged, the aetiology is most often central, but FMPP must also be considered

Simplified diagnostic algorithm for precocious puberty in girls



Nakamoto JM, Franklin SL and Geffner ME. Pediatric Practice Endocrinology. McGraw-Hill, 2010, p 271.

Simplified diagnostic algorithm for precocious puberty in boys



Nakamoto JM, Franklin SL and Geffner ME. Pediatric Practice Endocrinology. McGraw-Hill, 2010, p 271.

Treatment of precocious puberty

Depending on whether the child has:

- Central precocious puberty
- Peripheral precocious puberty

Treatment of **central** precocious puberty

- Consensus statement from the European Society for Pediatric Endocrinology and the Pediatric Endocrine Society highlighted that reasons for treatment include:
  - Preservation of adult height potential
  - Psychological difficulties with early puberty and menarche

Carel JC et al. Pediatrics 2009;123:e752-762.

Treatment of **central** precocious puberty

- Not all cases of CPP requires treatment
- Mainstay of treatment is GnRH analogues
- Provides constant serum levels of GnRH activity and thus overrides the pulsatility of endogenous GnRH
- Usual depot IM injection of leuprolide acetate once every 4 weekly

Carel JC et al. Pediatrics 2009;123:e752-762.

Treatment of **central** precocious puberty

- Adult height preservation particularly in girls <6 years with the greatest ⇧ in adult height after treatment
- Those between 6 and 8 years form a more heterogeneous group
- Some with very rapid or advanced puberty may benefit
- Many with slowly progressive puberty yield a normal height outcome without treatment
- Little utility to treat girls over age 8 years for the purpose of increasing adult height

Carel JC et al. Pediatrics 2009;123:e752-762.

Treatment of **peripheral** precocious puberty

- Treatment depends on the cause
  - Surgical removal of tumours ± chemotherapy and/or radiation therapy as indicated
  - Large ovarian cysts: drainage under USG guidance
  - Exogenous oestrogens or androgens: should be uncovered and eliminated
  - Non-classical CAH: exogenous glucocorticoid
- GnRHa therapy may be needed in addition to any of the above therapies if central puberty concomitantly starts at an early age

Thank you